SAS® Programming 1: Essentials Case Study

Course Notes

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Lesson 1 United States Airport Claims

1.1 Case Study Introduction

In this case study, you solve a real-world business problem by applying concepts that you learned in the SAS Programming 1: Essentials course. Be aware that there are numerous solutions to this problem, and some can include concepts that are outside the scope of the SAS Programming 1 course.

Background Information

You are a SAS programmer with six months of experience who is in charge of creating basic reports and maintaining SAS programs. Recently, you completed a SAS Programming course, and your supervisor gives you your first SAS programming project.

Business Problem

Your first project is to prepare and analyze Transportation Security Administration (TSA) Airport Claims data from 2002 through 2017. The TSA is an agency of the United States Department of Homeland Security that has authority over the security of the traveling public. A claim is filed if you are injured or your property is lost or damaged during the screening process at an airport.

To complete your project, you follow your supervisor's requirements, which are in Section 1.3 of this document. Here is what you need to do:

- Prepare the data.
- Create one PDF report that analyzes the overall data as well as the data for a dynamically specified state.

Data Information

The data that you use is **TSAClaims2002_2017.csv**, which was created from the following:

- TSA Airport Claims data from https://www.dhs.gov/tsa-claims-data.
- FAA Airport Facilities data from https://www.faa.gov/airports/airport_safety/airportdata_5010/.

The **TSAClaims2002_2017.csv** file was created by concatenating each individual TSA Airport Claims table. After the concatenation, the data was joined with the FAA Airport Facilities data. Here are a few notes regarding the data:

- All data is public data, and accuracy is not guaranteed.
- The column **Airport_Codes** from the TSA Airport Claims data has been joined with **Location_ID** from the FAA Airports Facilities data. Some **Airport_Codes** values do not correspond to **Location_ID** values.
- Columns in the TSA Airport Claims data have changed over the years. Because of this, some of the original columns were removed from the data for this case study.
- The column **Item_Category** does not have consistent input values over the years. For this reason, you do not clean this column in this case study.

Resources

To attempt this case study, you need to download the **TSAClaims2002_2017.csv** file.

1.2 Data Layout

Here is the column information for the **TSAClaims2002_2017.csv** table.

Column	Description
Claim_Number	Number for each claim. Some claims can have duplicate claim numbers but different information for each claim. Those claims are considered valid for this case study. Any duplicate rows should be removed from the data.
Date_Received	Date the claim was received. **Date_Received** must occur after **Incident_Date**. Range: From 2002 through 2017
Incident_Date	Date the incident occurred. **Incident_Date** must occur before **Date_Received**. Range: From 2002 through 2017
Airport_Code	Airport code three-letter abbreviation.
Airport_Name	Full name of the airport.
Claim_Type	Category of the claim. If the claim is separated into two types by a slash, **Claim_Type** is the first type. For example: Personal Property Loss/Injury is considered Personal Property Loss. Possible values (14): • Bus Terminal • Complaint • Compliment • Employee Loss (MPCECA) • Missed Flight • Motor Vehicle • Not Provided • Passenger Property Loss • Passenger Theft • Personal Injury • Property Damage • Property Loss • Unknown • Wrongful Death

Column	Description
Claim_Site	Airport location of the claim. Possible values (8): • Bus Station • Checked Baggage • Checkpoint • Motor Vehicle • Not Provided • Other • Pre-Check • Unknown
Item_Category	Type of items that have been filed in the claim. Depending on the year of the data, the **Item_Category** values are input differently. Because of varying consistency, you do not clean this column for the case study.
Close_Amount	The dollar amount a claim was closed for.
Disposition	The final settlement of the claim. Possible values (10): • *Insufficient • Approve in Full • Closed:Canceled • Closed:Contractor Claim • Deny • In Review • Pending Payment • Received • Settle • Unknown *Insufficient is the value from the raw data.
StateName	Associated airport state name (for example, NEW YORK). Requirements: Values should be in all proper case. (Original data is in all uppercase.)
State	Associated airport state code. This is the standard two-letter abbreviation used by the post office for US states and territories (for example, IL, PR, CQ). Requirements: Values should be in all uppercase.
County	Airport associated county (or parish) name (for example, Cook).
City	Associated airport city name (for example, CHICAGO).

1.3 Requirements

To prepare and analyze the data, you follow the requirements given to you by your supervisor. Be aware that these requirements are only assumed for this case study. They are not an accurate representation of TSA requirements.

Data Requirements

- Import the raw data file **TSAClaims2002_2017.csv**.
- The final data should be in the permanent library **tsa**, and the data set should be named **claims_cleaned**.
- Entirely duplicated records need to be removed from the data set.
- All missing and "-" values in the columns **Claim_Type**, **Claim_Site**, and **Disposition** must be changed to *Unknown*.
- Values in the columns **Claim_Type**, **Claim_Site**, and **Disposition** must follow the requirements in the data layout.
- All **StateName** values should be in proper case.
- All **State** values should be in uppercase.
- You create a new column named **Date_Issues** with a value of *Needs Review* to indicate that a row has a date issue. Date issues consist of the following:
  - a missing value for **Incident_Date** or **Date_Received**
  - an **Incident_Date** or **Date_Received** value out of the predefined year range of 2002 through 2017
  - an **Incident_Date** value that occurs after the **Date_Received** value
- Remove the **County** and **City** columns.
- Currency should be permanently formatted with a dollar sign and include two decimal points.
- All dates should be permanently formatted in the style 01JAN2000.
- Permanent labels should be assigned columns by replacing the underscores with a space.
- Final data should be sorted in ascending order by **Incident_Date**.

Report Requirements

The final single PDF report needs to ***exclude all rows with date issues*** in the analysis and answer the following questions:

1. How many date issues are in the overall data?
2. How many claims per year of **Incident_Date** are in the overall data? Be sure to include a plot.
3. Lastly, a user should be able to dynamically input a specific state value and answer the following:
   a. What are the frequency values for **Claim_Type** for the selected state?
   b. What are the frequency values for **Claim_Site** for the selected state?
   c. What are the frequency values for **Disposition** for the selected state?
   d. What is the mean, minimum, maximum, and sum of **Close_Amount** for the selected state? Round to the nearest integer.

1.4 Assignment Guide

Below is a suggested guide to help you solve the business problem. Be aware that there are multiple solutions to this problem and that you do not need to follow the steps below.

You can follow the SAS programming process steps below to solve the business problem. How you solve each requirement is your choice, but if you are stuck, you can refer to the **Hints** section in this document or post a question in the discussion forums.

Access Data

1. Import the **TSAClaims2002_2017.csv** file.

Explore Data

1. Preview the data.
2. Explore the following columns and make note of any adjustments needed using the information from the **Data Layout** and **Requirements** sections above.
   a. **Claim_Site**
   b. **Disposition**
   c. **Claim_Type**
   d. **Date_Received**
   e. **Incident_Date**

Prepare Data

1. Remove duplicate rows.
2. Sort the data by ascending **Incident_Date**.
3. Clean the **Claim_Site** column.
4. Clean the **Disposition** column.
5. Clean the **Claim_Type** column.
6. Convert all **State** values to uppercase and all **StateName** values to proper case.
7. Create a new column to indicate date issues.
8. Add permanent labels and formats.
9. Drop **County** and **City**.

Analyze Data

1. Analyze the overall data to answer the business questions. Be sure to add appropriate titles.
2. Analyze the state-level data to answer the business questions. Be sure to add appropriate titles.

Export Data

1. Export the end results into a single PDF named **ClaimReports** that has a style of your choice.

2. Customize the procedure labels in your report.

1.5 Results

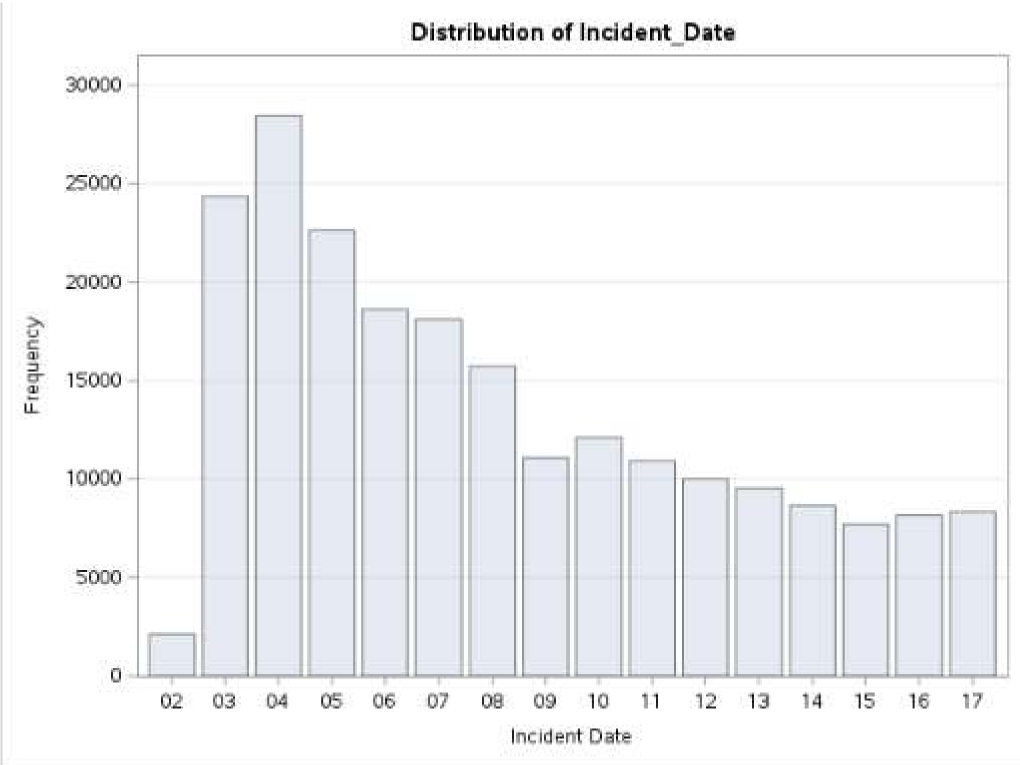
Here are the results for the overall analysis and a report with the selected state of California.

1. How many date issues are in the overall data?

Data Issues				
Date_Issues	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Needs Review	4241	100.00	4241	100.00
Frequency Missing = 216609				

2. How many claims per year of **Incident_Date** are in the overall data? Be sure to include a plot.

Incident Date				
Incident_Date	Frequency	Percent	Cumulative Frequency	Cumulative Percent
02	2123	0.98	2123	0.98
03	24359	11.25	26482	12.23
04	28484	13.15	54966	25.38
05	22631	10.45	77597	35.82
06	18643	8.61	96240	44.43
07	18116	8.36	114356	52.79
08	15727	7.26	130083	60.05
09	11075	5.11	141158	65.17
10	12108	5.59	153266	70.76
11	10921	5.04	164187	75.80
12	9984	4.61	174171	80.41
13	9536	4.40	183707	84.81
14	8659	4.00	192366	88.81
15	7721	3.56	200087	92.37
16	8182	3.78	208269	96.15
17	8340	3.85	216609	100.00



3. Dynamically input a specific state value and answer the following:
   a. What are the frequency values for **Claim_Type** for the selected state?

Claim Type				
Claim_Type	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Complaint	7	0.03	7	0.03
Compliment	2	0.01	9	0.04
Employee Loss (MPCECA)	51	0.20	60	0.24
Missed Flight	1	0.00	61	0.24
Motor Vehicle	17	0.07	78	0.31
Passenger Property Loss	14892	59.63	14970	59.95
Passenger Theft	55	0.22	15025	60.17
Personal Injury	194	0.78	15219	60.94
Property Damage	8996	36.02	24215	96.97
Property Loss	1	0.00	24216	96.97
Unknown	756	3.03	24972	100.00

   b. What are the frequency values for **Claim_Site** for the selected state?

Claim Site				
Claim_Site	Frequency	Percent	Cumulative Frequency	Cumulative Percent
Bus Station	1	0.00	1	0.00
Checked Baggage	19553	78.30	19554	78.30
Checkpoint	5142	20.59	24696	98.89
Motor Vehicle	19	0.08	24715	98.97
Other	187	0.75	24902	99.72
Pre-Check	2	0.01	24904	99.73
Unknown	68	0.27	24972	100.00

   c. What are the frequency values for **Disposition** for the selected state?

Disposition	Frequency	Percent	Cumulative Frequency	Cumulative Percent
*Insufficient	187	0.75	187	0.75
Approve in Full	5266	21.09	5453	21.84
Closed:Canceled	49	0.20	5502	22.03
Closed:Contractor Claim	83	0.33	5585	22.37
Deny	10918	43.72	16503	66.09
In Review	1086	4.35	17589	70.43
Received	3	0.01	17592	70.45
Settle	4192	16.79	21784	87.23
Unknown	3188	12.77	24972	100.00

   d. What is the mean, minimum, maximum, and sum of **Close_Amount** for the selected state? Round to the nearest integer.

California: Close_Amount Statistics

The MEANS Procedure

Analysis Variable : Close_Amount Close Amount			
Mean	Minimum	Maximum	Sum
98	0	14519	2096386